Fiscal Year:	FY 2018	**Task Last Updated:**	FY 04/03/2018
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Project Title:	Complex Plasma under Microgravity, Utilizing the International Space Station Experiment “PK-4” and Beyond		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	04/15/2017	**End Date:**	04/15/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thomas, Edward Ph.D. (Auburn University) Williams, Jeremiah Ph.D. (Wittenberg University)		
Grant/Contract No.:	JPL 1571699		
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Task Description:	Complex plasmas are four-component systems that contain electrons, ions, neutral atoms, and charged microparticles (i.e., the dust particles). Depending upon the experimental conditions, the microparticles in these systems can show solid-like, liquid-like, or gas-like properties. However, ground-based experiments are subject to gravitational forces that often dominate the behavior of the systems under investigation leading to sedimentation and anisotropic effects. Therefore, in order to study large-scale complex plasmas and the influences of the weaker forces that are masked by the gravitational force, experiments have to be performed under “microgravity” conditions. Over the past decade, several complex plasma experiments have been performed using the International Space Station (ISS). The latest of these research instruments, PK-4, is the first microgravity laboratory that is optimized to study fluid phenomena and fundamental effects such as particle charging, screening, and the influence of a directed ion flow. Additionally, since PK-4 is a multi-user facility, we have a unique opportunity to use this instrument to study ion/particle and particle/particle interactions that we otherwise would not be able to address in ground-based experiments. The intellectual merit of this research project is centered on three major research topics. Topic 1: Charging and interaction of microparticles in the plasma. The most important measurement in a complex plasma is the measurement of the charge on the particles. However, direct experimental measurement of the charge is difficult, because it usually relies on additional assumptions about experimental parameters such as the plasma electric field and on the validity of an assumed particle interaction type. The microgravity environment of PK-4, where the dust is suspended in the bulk plasma, allows experiments that can potentially lead to a higher accuracy in the particle charge determination. Topic 2: Ion-dust interactions and the thermal properties of dusty plasmas. A key observation in complex plasmas is the unusually high dust kinetic temperature. It has been postulated that ion-dust interactions are largely responsible for heating the dust particles. The microgravity environment of the PK-4 laboratory, in combination with the tools to actively manipulate the ions in the plasma, provides a unique platform for studying the transfer of energy between ions and the microparticles in the plasma. Topic 3: Particle transport and heating in waves. The low charge to mass ratio of the dust leads to characteristic frequencies for the dust component that are on the order of a few Hz. This, coupled with the finite size of the dust grains, allows researchers a unique opportunity to directly image plasma physics phenomena and particle transport at the kinetic level. In the microgravity environment of PK-4, it will be possible to measure the transport properties of particles during their initial injection into the bulk plasma, follow the particles as they form collective structures (e.g., waves), and study how ion flow modifies the thermal properties of these waves. This proposed research project has several opportunities to have a broader impact: (1) through on- and off-campus outreach activities and the training of undergraduate and graduate students, this project will cultivate the next generation of plasma scientists; (2) through the development of new ground-based laboratory instruments (e.g., plasma sources, diagnostics) that will complement the proposed research and contribute to the development of next-generation dusty plasma microgravity experiments; and (3) through the development of new, advanced software tools for image analysis and particle tracking that will support the proposed studies and be offered as community-wide research tools.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research performed under this project falls in the main category of fundamental research. Often, findings obtained by performing fundamental research show their true impact for the society only after many years. However, the study of dusty/complex plasma can benefit the overall research infrastructure in many ways already in the timely vicinity of any discoveries made, mainly as a result of the strong interdisciplinary nature of this research. Studying complex plasma topics such as wave phenomena, heating, the spatial separation of phases, their interaction or the transition between phases such as from fluids to crystals all being analyzed down to the scale of individual particles can lead to fundamental findings that might be immediately applied to many other research fields such as solid state physics, fluid physics, plasma physics, astrophysics. As the specific research performed within the scope of this project makes use of the microgravity (free-fall) environment supported by the national laboratory in space, the International Space Station (ISS), further utilizing an experiment that is build by the European Space Agency (ESA), highly international cooperation activities are a direct result. It is likely that the effort to work on the same fundamental questions in a cooperative acting international team will ultimately benefit each involved nation also beyond just the research. Some aspects of the specific research topics within the scope of this project involve the development of universal software, hardware or analysis tools that can also be used under different research situations, and some of the more specific technical tools that are developed, such as a plasma impedance probe, might even help to overall improve technology that can be used for industrial purposes. Finally, as part of the general research mission, we train the next generation of engineers and scientists who can in the course of their future career produce a large benefit for the overall economic situation of the nation and beyond.
	The project started on 04/15/2017. The majority of the workforce at Auburn University was already in place at this time. The workforce at Wittenberg University needed some additional time to be fully developed as it depended on the realization of a subcontract formalism between both involved universities. The full manpower was reached with the start of the fall semester when all major undergraduate and graduate research assistants had been hired. Additional support was granted from visiting research assistants and other undergraduate/graduate students who were not paid by the grant. To archive the scientific goals the project also made wide use of, not only but mainly, a flourishing cooperation with the main team members of the leading core group of the International Space Station (ISS) Experiment "Plasma Kristall 4" (PK-4) localized at the German Aerospace Center (DLR e.V.) in Wessling, Germany. The progress described in this report is for the first year of the project where, as anticipated in the project planing, many tools (software, hardware, methods) have to be developed to study the proposed topics. As such, it is expected to have some higher weight towards development activities and will show less results as publications. Further, the ISS experiments PK-4, which will be utilized for this project, were not operated in 2017 and early 2018 as a result of a technical issue, which was studied and is on the way to be fixed in the first half of 2018. Latter instance interfered to a minor extent with the planned activities of the here reported project. It was taken account for by shifting some activities in their timeline and by utilizing the PK-4 ground modules as experimental preparation tool not only for the planned space experiment but also to derive the required analysis tools and study the covered physics also under gravitational influence.

Task Progress:

The mission of the described research project is dedicated to gain new knowledge in the physics of complex (dusty) plasmas and to create a vision beyond the ongoing experiments towards future space experiments on complex plasmas in space (presumably also aboard the ISS). The research that is conducted is categorized in the major three research topics:

Topic T1: Charging and interaction of micro-particles in a plasma, Topic T2: Ion-dust interaction and the thermal properties of dusty plasmas, Topic T3: Particle transport and heating in waves,

and it also contains activities that go beyond the immediate research by utilizing the PK-4 space- and ground-based experiment modules. Breadboard experimental setups, which are in the planning phase as successor experiments to PK-4, are used to investigate topics beyond the original scope. Still, most of these activities have a clear relation to the major research topics T1-T3 and are thus described under the most fitting category, while activities purely related to the development of next generation experiments or those that are related to PK-4 administrative work of the PIs are summarized under the topic:

Topic TX: Future space experiments, administrative tasks, and analysis tool developments.

In the near future, some experiments dedicated to the scientific topics have been and will be carried out locally at Auburn University and Wittenberg University, while the experiments that are planned to be performed aboard the ISS by utilizing the PK-4 experiment setup, are defined under close logistical support by the DLR group. Using the established manpower and opportunities given by the local laboratories and the close cooperation with the German Aerospace Center (DLR e.V), we were able to archive the following described progress towards our project goals:

Topic T1: A major goal of this project is to investigate the charging of the dust particles and their mutual interaction in the plasma. There are multiple ways that can be used to derive an appropriate experimentally determined estimate for the particle charge, for example, by investigating dust-dust particle collisions, studying normal/wave modes in the system of many particles or, via an indirect approach, by determining the discrepancy of electron to ion density that is balanced by the dust particle density times the charge number on each particle.

In the first year of this project we followed two major approaches--a direct one that is based on dust particle collisions and the indirect one, which is based on the determination of the plasma densities in a dusty plasma. Two graduate research assistance (GRA) students were working on these topics, one 100% and the other an average of 50%. Determining the effective particle charge from collision is a straight forward task, if the involved particles are not surrounded by other particles and if the collision takes place in the right geometry embedded in a homogeneous background plasma. Since this is not the typical case in a PK-4 based experiments where millions of particle float in close vicinity to each other while subject to a directed ion flow, the first year activities were planned to first concentrate on setting up a benchmarking system. As part of this project, one student started working on a molecular dynamic benchmarking system that, when completed, will allow to generate synthetic data for which the underlying individual interaction and charges are known. Thus it can be used to benchmark/calibrate further analysis tools. Toward the second half of the project, the derived and calibrated dust collision analysis tools will be applied to experimental data if the benchmark will demonstrate its successful operation. At the end of year one the GRA was able to establish a working, basic simulation tool that allows to simulate the interaction and dynamics in a two dimensional system for multiple thousands of particles as are expected for a real experiments. The progress is about what could be expected and what was planned in the project proposal for year one on this topic.

The second GRA student is working on an indirect method to determine the particle charge. He is developing a probe system based on measuring the electrical ringing in the plasma (plasma resonances), measuring the plasma conductivity and using standard plasma probe tools such as a Langmuir probe in combination to estimate the discrepancy in the electron and ion densities in the pure or dust enriched plasma. The discrepancy, theoretically, has to be balanced by the charge on each particle multiplied by the dust number density which can easily be determined just from camera images of illuminated dust particle clouds. This "simple" balance can be used to estimate the dust particle charge, but it is a complex challenge to realize an adequate, precise method to determine electron and ion densities.

During year one, the student working on this topic constructed several kinds of plasma probes mainly to investigate the plasma resonances (determining the electron density) and to perform basic Langmuir probe measurements (determining electron and ion densities). He tested these probes in plasma with and without a dust component under a variety of situation changing plasma power, pressure and thus density. The first, recently obtained data show clearly a substantial effect of the dust component on the resonance frequencies in the plasma and thus at least the electron density (electron plasma frequency). A complete set of measurements where electron, ion, and dust densities are obtained simultaneously for the same plasma conditions is not yet archived, but is part of the second year effort. The development of this plasma diagnostic tool shows a lot of potential to reach beyond this specific task as it might be developed to a standard plasma diagnostic tool. We hope in the future also to adapt it in a way that it can be applied as an in-situ diagnostic, performing a measurement in an ongoing experiment without causing any disturbances to the experimental conditions. Latter will be extremely useful, especially for next generation complex/dusty plasma experiments for the ISS.

Topics T2/T3: Topics T2 and T3 are performed at a reduced scale compared to the initial proposal as the granted funds were much lower than requested and thus under a reduced manpower. As such we report on the effort on these two topics combined. Both Co-Investigators (CoIs) work as the leading team on these research topics supported full-time since early 2018 and before that part time by a GRA (financed from other sources) and also making substantial use of collaborative support by the core group of the PK-4 complex plasma ISS experiment.

Based on earlier provided data from the PK-4 core team our group was able already from the beginning to perform some analysis on real PK-4 ISS based data. We were looking for the effects that the polarity switching of the DC-discharge (PK-4) induce to the dust cloud in the plasma, specifically how the polarity switching can lead to heating in the dusty plasma. The analysis of such experiments is based on the determination of the velocity distribution of particles, while the discharge is switched in its polarity back and forth. The analysis of the early obtained data showed an instant dust cloud temperature rise as a result of the igniting of the polarity switching mode. To investigate this effect in greater detail and to determine what the exact origin of this heating might be, dedicated experiments were proposed. As PK-4 was not operational during the main part of 2017, because of a minor technical issue, the dedicated experiments were performed in a first run, supported by the PK-4 core group, at one of the ground-based PK-4 modules at DLR. These experiments were performed by our team members when visiting DLR for additional, other activities. It is planned to perform follow-up experiments also aboard the ISS when PK-4 will be back on-line in 2018. A GRA is now working on setting up a molecular dynamic simulation model for the same situation to investigate the extend and origin of this

instantaneous heating effect. One of the Co-Is is also working in cooperation with DLR on the definition of dedicated experiments both for the PK-4 ground module and the ISS counterpart related to particle transport and heat transport via waves in the dusty plasma. First experiments have been performed in December and are under investigation now.

Topic TX: In addition to the activities described above, the Principal Investigator (PI) performed two trips to the German Aerospace Center in Germany during 2017 to support the development of the next generation dusty/complex plasma space experiment "EkoPlasma," formerly called "PlasmaLab." As part of both visits the opportunity was taken also to perform new kinds of experiments to realize a phase transition in the state of a complex plasma by means of enhancing the particle coupling strength via compression. This is similar as gases can liquefy under compression. These experiments were possible and are so far to our knowledge unique as the new complex plasma device, the heart of EkoPlasma, supports changing the internal plasma confinement by moving the electrodes towards each other and thus reducing the plasma volume in an ongoing experiments. The obtained data show clearly that a liquid complex plasma can crystallize if it is actively compressed. The data is under detailed analysis now but requires further development work on the PI's image analysis tools. Some effort was taken here in 2017, but is expected to substantial continue into 2018 (year two of the project). Also the other Co-Is of the project visited DLR multiple times, mainly to perform experiments related to topics T2/T3 but also in case of the second visit to participate in the Facility Science Team (FST) meeting that took place in Germany at DLR, December 2017. At this meeting the status of the scientific mission of PK-4 was discussed and recommendations given how to proceed. The FST is in general overseeing the scientific mission of the PK-4 space experiment.

Besides performing the individual activities, we already had the chance on multiple occasions to share our mission, the scientifically gained knowledge and the related developments with the complex/dusty plasma as well as the general plasma physics communities. Multiple presentations were given by the PI/Co-Is and the students on two major conferences such as the 8th International Conference on the Physics of Dusty Plasma in Prague, Czech Republic, as well as the 59th Annual American Physical Society Meeting of the Division of Plasma Physics in Milwaukee, WI, USA. Abstracts submitted to latter are publicly available on-line. Additional presentations by the PI related to the project were given during the Annual NASA Fundamental Physics Workshop in Santa Barbara, CA, USA. In summary, we believe the first year of our project was very successful, which is also indicated by the number of presentations given by the team members of this project during the reporting period.

Bibliography Type:	Description: (Last Updated: 09/30/2025)
Abstracts for Journals and Proceedings	Lynch B, Konopka U, Funk D, Thomas E. "Microparticle Deflection in a Plasma As a Result of a Strong Magnetic Field." Presented at the 59th Annual Meeting of the APS Division of Plasma Physics, Milwaukee, WI, October 23-27, 2017. Bulletin of the American Physical Society. 2017;62(12):Abstract ID: BAPS.2017.DPP.CP11.5. http://meetings.aps.org/link/BAPS.2017.DPP.CP11.5 , Oct-2017
Abstracts for Journals and Proceedings	Funk D, Konopka U, Thomas E. "Molecular Dynamic Simulation of Weakly Magnetized Complex Plasmas." Presented at the 59th Annual Meeting of the APS Division of Plasma Physics, Milwaukee, WI, October 23-27, 2017. Bulletin of the American Physical Society. 2017;62(12):Abstract ID: BAPS.2017.DPP.CP11.12. http://meetings.aps.org/link/BAPS.2017.DPP.CP11.12 , Oct-2017
Abstracts for Journals and Proceedings	Doyle B, Konopka U, Thomas E. "RF attenuation as a dusty plasma diagnostic." Presented at the 59th Annual Meeting of the APS Division of Plasma Physics, Milwaukee, WI, October 23-27, 2017. Bulletin of the American Physical Society. 2017;62(12):Abstract ID: BAPS.2017.DPP.CP11.13. http://meetings.aps.org/link/BAPS.2017.DPP.CP11.13 , Oct-2017
Abstracts for Journals and Proceedings	Konopka U, Thomas E, Funk D, Doyle B, Williams J, Knapek C, Thomas H. "Complex Plasmas under free fall conditions aboard the International Space Station." Presented at the 59th Annual Meeting of the APS Division of Plasma Physics, Milwaukee, WI, October 23-27, 2017. Bulletin of the American Physical Society. 2017;62(12):Abstract ID: BAPS.2017.DPP.CP11.14. http://meetings.aps.org/link/BAPS.2017.DPP.CP11.14 , Oct-2017
Dissertations and Theses	Lynch B. "Micro-Particle Dynamics in Strongly Magnetized Low Temperature Plasmas." Dissertation, Auburn University, Auburn, Alabama, August 2017. , Aug-2017